



Comparison Of Papers

Title of the Paper	Problem	Objectives	Methodologies	Keywords	Results / Contributions
Blockchain Based Land Record Management System (IJRASET, 2025)	Centralized land record systems are vulnerable to document tampering, unauthorized modifications, duplicate registrations, and fraudulent land sales.	Create a secure, transparent, and tamper-proof platform to automate land registration, ownership verification, and property transfers.	Ethereum Blockchain, Solidity Smart Contracts, MetaMask Authentication, Cryptographic Hashing, React.js Frontend.	Blockchain, Land Registry, Smart Contracts, Solidity, E-Governance, Deed Verification.	Developed a tamper-resistant smart governance platform that automates property transfers and eliminates intermediary fraud.
A Blockchain-IPFS Framework for Secure Healthcare Data (SN Computer Science, 2025)	Storing large medical records directly on a blockchain causes high gas fees and latency, while traditional systems risk data breaches.	Build a scalable, privacy-preserving healthcare framework with off-chain record storage and dynamic doctor access control.	IPFS Content Identifiers (CIDs), Ethereum Smart Contracts, Role-Based Access Control (RBAC), AES-256 Encryption.	Blockchain, IPFS, Healthcare, Smart Contracts, EHR, Access Control, CID.	Reduced blockchain storage costs and gas fees by 16% through IPFS off-chain storage while maintaining patient-controlled access rights.
Blockchain Based Identity Management System (IJSRCSEIT, 2025)	Centralized identity databases suffer from single points of failure, frequent data breaches, and identity theft risks.	Implement a user-centric, decentralized identity management system to enable privacy-preserving digital authentication.	Decentralized Identifiers (DIDs), Verifiable Credentials, Smart Contracts, Cryptographic Proofs.	Blockchain, Identity Management, DIDs, Smart Contracts, Verifiable Credentials, Privacy.	Created a tamper-resistant identity verification framework that allows users to prove credentials without exposing raw data.

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Blockchain-Based Digital Identity Management Systems (IJACECT, 2025)	Web applications rely on passwords and centralized SQL databases, making user accounts vulnerable to leaks and unauthorized access.	Eliminate password-based authentication by implementing Web3 Role-Based Access Control (RBAC) across web portals.	Role-Based Access Control (RBAC), Smart Contract Mapping, MetaMask Signature Authentication, Distributed Ledger Technology.	Blockchain, Identity Management, Web3, RBAC, Wallet Authentication, Access Control.	Replaced traditional database login credentials with a dynamic permission portal that unlocks features based on wallet clearance.
A Blockchain-Based Framework for Academic Certificate Verification Using IPFS (WJARR, 2026)	Physical and standard digital certificates are easily forged using editing tools, and manual degree verification takes weeks.	Build an instant, public document verification portal for academic records using decentralized cryptographic hashing.	IPFS Content Identifiers (CIDs), Solidity Smart Contracts, Web3.js, Digital Signatures.	Blockchain, IPFS, Academic Certificates, Verification, Hash Proof, Tamper-Proof.	Designed an instant file-drop verification system that detects modified text or altered grades through cryptographic hashes.

<https://www.ijraset.com/research-paper/blockchain-based-land-record-management-system> {1}

<https://doi.org/10.1007/s42979-025-03936-z> {2}

https://www.researchgate.net/publication/389967371_Blockchain_Based_Identity_Management_System {3}

https://www.researchgate.net/publication/398237449_Blockchain-Based_Digital_Identity_Management_Systems {4}

<https://wjarr.com/content/blockchain-based-framework-academic-certificate-verification-using-ipfs> {5}

Term	Simple Meaning
Web3	The decentralized web where users control their own data and log in using digital wallets instead of usernames and passwords.
IPFS	A peer-to-peer storage network used to store large files (like PDFs) off-chain to avoid high blockchain storage costs.
CID	Content Identifier; a unique 46-character cryptographic fingerprint generated by IPFS for any uploaded file.
Smart Contract	Self-executing programs stored on the blockchain that run automatically when defined rules are met.
DID	Decentralized Identifier; a digital identity credential owned entirely by the user without relying on a central authority.
RBAC	Role-Based Access Control; restricting or granting access to specific web pages or data based on user roles (Admin, User, Doctor).
Testnet	A free, identical copy of a blockchain network used by developers to test applications without spending real money.

Technologies & Tools Used

- **Language:** Solidity, JavaScript, HTML/CSS
- **Frontend:** React.js / HTML5
- **Blockchain Environment:** Remix IDE, Ganache, Hardhat
- **Network:** Ethereum Sepolia Testnet
- **Web3 Integration:** Ethers.js, Web3.js
- **Wallet & Auth:** MetaMask Extension
- **Off-Chain Storage:** Pinata IPFS API